

illuminati - 2020

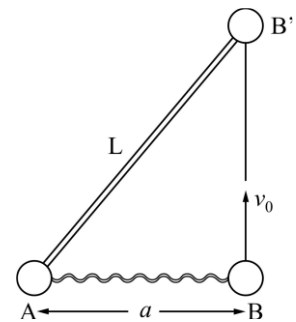
Advanced Physics Assignment-3

Vectors, Kinematics, 2D-Motion, Dynamics of A particle,
Energy and Momentum, Rotational Motion, Gravitation

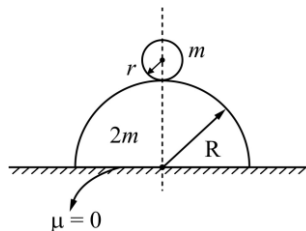
SECTION 1

SINGLE CORRECT ANSWER TYPE

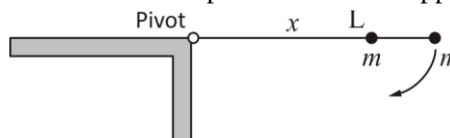
1. Two identical spheres A and B, each of mass m are attached to an inextensible inelastic cord of length L , and are resting at a distance a from each other on a frictionless horizontal surface. Sphere B is given a velocity v_0 in a direction perpendicular to line AB and moves straight until it reaches B' when the cord becomes taut. The energy lost as the cord becomes taut is given by :



- (A) $\frac{mv_0^2}{2L^2}(L^2 - a^2)$ (B) $\frac{mv_0^2}{4L^2}(L^2 - a^2)$
- (C) $\frac{mv_0^2}{L^2}(L^2 - a^2)$ (D) None of these
2. A small sphere of radius r and mass m is placed on a big hemisphere of mass $2m$ and radius R which is kept on a smooth horizontal surface as shown in the figure. Find the displacement of hemisphere when line joining between centre of small sphere and centre of big hemisphere makes 30° with the vertical. (initially this line was vertical, and assume small sphere is not losing the contact with hemisphere)

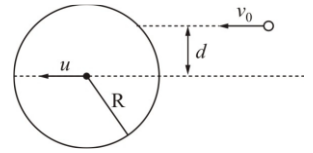


- (A) $\frac{R+r}{2}$ (B) $\frac{R+r}{3}$ (C) $\frac{R+r}{6}$ (D) None of these
3. A massless stick of length L is pivoted at one end and has a mass m attached to its other end. It is held in a horizontal position. Where should a second mass m be attached to the stick, so that the stick falls with maximum angular acceleration as possible when dropped ?



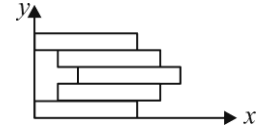
- (A) $x = \frac{L}{2}$ (B) $x = \frac{L}{2\sqrt{2}}$ (C) $x = (\sqrt{2} - 1)L$ (D) $x = \frac{3L}{4}$

4. A massive disc of radius R is moved with a constant velocity u on a frictionless table. Another small disc collides with it elastically with a speed of $v_0 = 0.3 \text{ m/s}$, the velocities of the discs are parallel. The distance d shown in the figure is equal to $\frac{R}{2}$, friction between the disc is

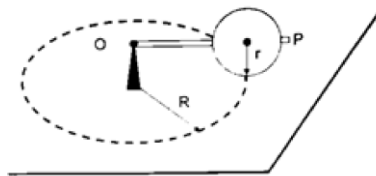


negligible. For which u (in m/s) will the small disc move perpendicularly to its original motion after the collision?

- (A) $\frac{1}{10}$ (B) $\frac{2}{10}$ (C) $\frac{3}{10}$ (D) $\frac{4}{10}$
5. Figure shows five uniform bricks each of length l . Each brick is placed over another with an offset of length $\frac{l}{10}$. Find the x -coordinate of centre of mass of this system.

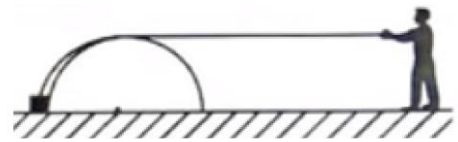


- (A) $2l$ (B) $\frac{29l}{50}$ (C) $1.5l$ (D) $0.5l$
6. A shell flying with velocity 500 m/s bursts into three identical fragments so that kinetic energy of the system increases 1.5 times. What maximum velocity can one of the fragments can attain?
- (A) 1000 m/s (B) 250 m/s (C) 500 m/s (D) 700 m/s
7. A ball moving in a straight-line collides elastically with another stationary ball of same mass such that at the moment of impact the angle between the line passing through balls centre is 45° with the direction of initial motion of striking ball. If balls are smooth, find maximum fraction of kinetic energy that transform to potential energy in process of collision.
- (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$
8. A steel ball of mass m falls from a height h on a horizontal surface of a table. Find the total momentum imparted by ball to table after numerous bounces, if every impact decreases the velocity of ball by n times ($n > 1$).
- (A) $m\sqrt{2gh} \left[\frac{n+1}{n-1} \right]$ (B) $m\sqrt{2gh} \left[\frac{n-1}{n+1} \right]$ (C) $m\sqrt{2gh} \left[\frac{2n+1}{2n-1} \right]$ (D) $m\sqrt{2gh} \left[\frac{2n-1}{2n+1} \right]$
9. A sphere of mass m and radius r rolls without sliding over a horizontal plane, rotating about a horizontal axis OP . During motion the centre of sphere moves at a velocity v along a circle of radius R . Find the kinetic energy of the sphere.



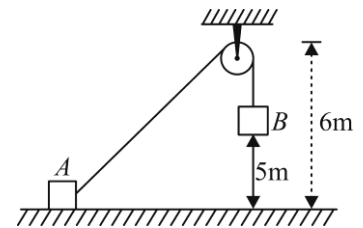
- (A) $\frac{7}{10}mv^2$ (B) $\frac{2}{5}mv^2 \left(1 + \frac{2}{7} \frac{r^2}{R^2} \right)$
- (C) $\frac{7}{10}mv^2 \left(1 + \frac{2}{7} \frac{r^2}{R^2} \right)$ (D) $\frac{7}{10}mv^2 \left(1 + \frac{7}{2} \frac{r^2}{R^2} \right)$

10. Figure shows a fixed hemisphere of radius R and person pulls a small body of mass m slowly by a string along the rough surface of hemisphere from ground to top of hemisphere. If the friction coefficient is μ , find the work done by the man in the process.



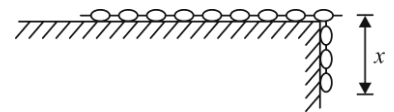
(A) $(\mu+1)mgR$ (B) mgR (C) $(\mu+2)mgR$ (D) μmgR

11. A block A of mass m is held at rest on a smooth horizontal floor. A smooth light pulley is fixed at a high 6 m above the ground over which a string of length 16 m connected to A as shown in figure. On the other end of string a block B of same mass m is connected and hanging at a height of 5 m above floor. If system is released, find the speed at which block B hits the floor.



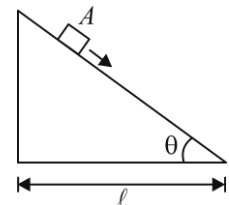
(A) 10 m/s (B) $v = \frac{40}{\sqrt{41}}$ m/s (C) $v = \frac{20}{\sqrt{41}}$ m/s (D) $v = \frac{40}{\sqrt{82}}$ m/s

12. A chain of mass m and length l rests on a rough table with one end is hanging over the edge as shown. The friction on table is such that chain starts sliding all by itself when $x = l/3$. Find the total work done by friction on chain till chain completely slides off.



(A) $-\frac{1}{3}mg\ell$ (B) $-\frac{1}{6}mg\ell$ (C) $-\frac{1}{9}mg\ell$ (D) $-\frac{1}{12}mg\ell$

13. Figure shows a fixed wedge block of base length l and inclination θ to horizontal. A small body A is released from top of inclined surface on which friction coefficient is k . Find the value of θ for which the time of sliding on inclined plane for body A will be minimum.



(A) $\theta = \frac{1}{2} \tan^{-1}(-k)$ (B) $\theta = \frac{1}{2} \sin^{-1}(-k)$
 (C) $\theta = \frac{1}{2} \cos^{-1}(-k)$ (D) $\theta = \frac{1}{2} \cot^{-1}(-k)$

14. A small satellite of mass ' m ' is revolving around earth in a circular orbit of radius r_0 with speed v_0 . At certain point of its orbit, the direction of motion of satellite is suddenly changed by angle $\theta = \cos^{-1}\left(\frac{3}{5}\right)$ by turning its velocity vector in the same plane of motion, such that speed remains constant. The satellite, consequently goes to elliptical orbit around earth. The ratio of speed at perigee to speed at apogee is :

(A) 3 (B) 9 (C) $\frac{1}{3}$ (D) $\frac{1}{9}$

15. In older times, people used to think that the Earth was flat. Imagine that the Earth is indeed not a sphere of radius R , but an infinite plate of thickness H . What value of H is needed to allow the same gravitational acceleration to be experienced as on the surface of the actual Earth? (Assume that the Earth's density is uniform and equal in the two models.)

(A) $\frac{2R}{3}$ (B) $\frac{4R}{3}$ (C) $\frac{8R}{3}$ (D) $\frac{R}{3}$

SECTION 2

LINK-COMPREHENSION TYPE

PARAGRAPH FOR QUESTIONS 16 - 18

A particle starts moving at $t = 0$ in x - y plane such that its coordinates (mm) change with time (in sec) as $x = 2t$ and $y = 10\sin(2t)$.

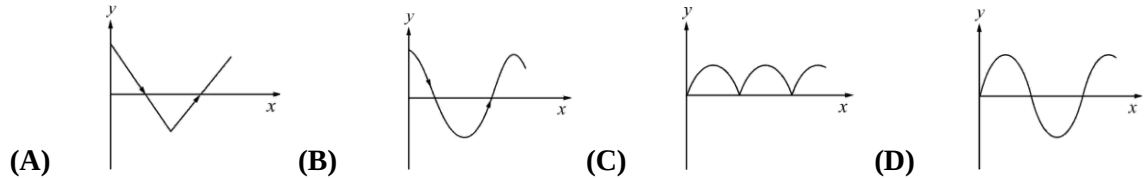
16. If magnitude of its acceleration a , then :

- (A) $a \propto x$ (B) $a \propto y$ (C) $a \propto \sqrt{x^2 + y^2}$ (D) $a = 0$

17. Maximum speed of the particle is :

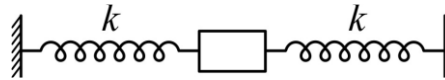
- (A) $12m/s$ (B) $2\sqrt{26}m/s$ (C) $2\sqrt{101}m/s$ (D) $10m/s$

18. The path of the particle will be :



PARAGRAPH FOR QUESTIONS 19 - 20

A block is tied within two springs, each having spring constant equal to k . Initially the springs are in their natural length and horizontal as shown in the figure, the block is released from rest. The springs are ideal, acceleration due to gravity is g downwards. Air resistance is to be neglect. The natural length of spring is l_0 .



19. If the decrease in height of the block till it reaches equilibrium is $\sqrt{3}l_0$ then the mass of the block is :

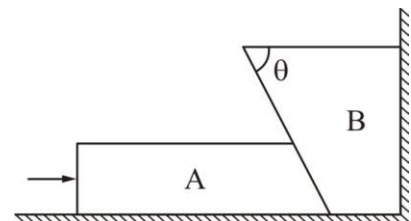
- (A) $\frac{2kl_0}{g}$ (B) $\frac{\sqrt{2}kl_0}{g}$ (C) $\frac{\sqrt{3}kl_0}{g}$ (D) None of these

20. When the block is under equilibrium as per the conditions of previous question, then the speed of the block is :

- (A) Zero (B) $\sqrt{\frac{4gl_0}{\sqrt{3}}}$ (C) $\frac{\sqrt{3}kl_0}{g}$ (D) None of these

PARAGRAPH FOR QUESTIONS 21 - 22

In the figure shown, the coefficient of friction between B and the wall is $\frac{2}{3}$ and the coefficient of kinetic friction between B and the wall is $\frac{1}{3}$. Other contacts are smooth. Mass of A is $2m$ and the mass of B is m . (Take $\tan \theta = \frac{3}{4}$.)



21. Find the minimum force F required to lift B , up.

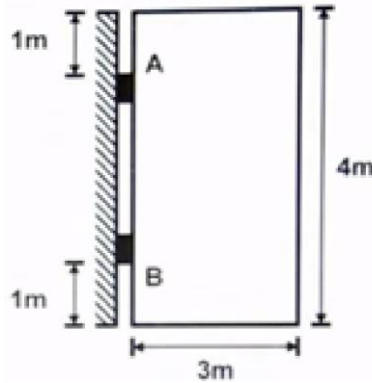
- (A) mg (B) $\frac{3}{2}mg$ (C) $\frac{2}{3}mg$ (D) $2mg$

22. If the force applied on A is slightly increased than the calculated value of minimum force, then find the acceleration of B .

- (A) $\frac{g}{22}$ (B) $\frac{g}{11}$ (C) $\frac{3g}{22}$ (D) $\frac{2g}{11}$

PARAGRAPH FOR QUESTIONS 23 - 25

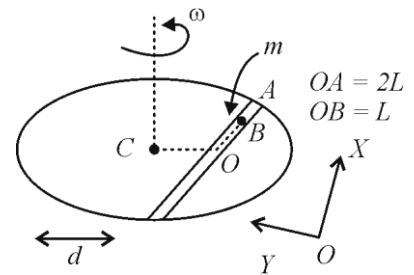
A door of 100 kg mass which is supported by two hinges A and B on a wall is of size 4m × 3m as shown. The force by hinges on door are equal.



23. Find the vertical normal reaction at hinge A.
 (A) 250 N (B) 500 N (C) 750 N (D) 1000 N
24. Find the horizontal normal reaction at hinge B.
 (A) 500 N, Right (B) 750 N, Right (C) 500 N, Left (D) 750 N, Left
25. Find the horizontal normal reaction at hinge A.
 (A) 500 N, Right (B) 750 N, Right (C) 500 N, Left (D) 750 N, Left

PARAGRAPH FOR QUESTIONS 26- 27

A frame of reference that is accelerated w.r.t. an inertial frame of reference is called a non-inertial frame of reference. A coordinate system fixed on a circular disc rotating about a fixed axis with constant angular velocity ω is an example of non-inertial frame and reference the relationship between the force $\vec{F}_{rot}$ experienced by a particle of mass m moving on rotating disc and the force $\vec{F}_{in}$ experienced by particle in an inertial frame is $\vec{F}_{rot} = \vec{F}_{in} + 2m \vec{V}_{rot} \times \vec{\omega} + m(\vec{\omega} \times \vec{r}) \times \vec{\omega}$.

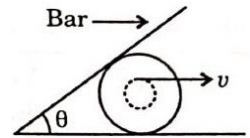


Where $\vec{V}_{rot}$ is velocity of particle in rotating frame of reference and $\vec{r}$ is position vector w.r.t. center and disc. Now consider a smooth slot along a chord at a distance 'd' from the centre of a disc rotating counter clockwise with const. Angular velocity ω about its vertical axis through center. We assign x axis along chord with origin at middle of chord. A small block & man m is gently placed at $t=0$ at $x=L$ and is constraint to move along slot only.

26. The distance r of particle at time t is :
 (A) $\frac{L}{2}(e^{\omega t} + e^{-\omega t})$ (B) $L \cos \omega t$ (C) $\frac{L}{2}(e^{2\omega t} + e^{-2\omega t})$ (D) $L \cos 2\omega t$
27. Net reaction force by disc on particle is :
 (A) $m\omega^2 \left[d + L(e^{2\omega t} - e^{-2\omega t}) \right] \hat{j} + mg\hat{k}$ (B) $m\omega^2 (d + L \cos 2\omega t) \hat{j} + mg\hat{k}$
 (C) $m\omega^2 \left[d + L(e^{\omega t} - e^{-\omega t}) \right] \hat{j} + mg\hat{k}$ (D) $m\omega^2 (d + L \cos \omega t) \hat{j} + mg\hat{k}$

PARAGRAPH FOR QUESTIONS 28 - 29

A yo-yo is pulled by its string along a horizontal surface without slipping. The horizontal velocity of the end of the string remains equal to v . A smooth bar is pivoted as shown and remains supported by the yo-yo. The outer and the inner radii of the yo-yo are R and r , respectively.



28. The velocity of centre of mass of yo-yo is :

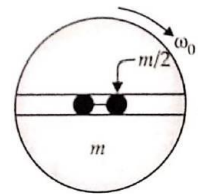
- (A) $\frac{vr}{R}$ (B) $\frac{v(R+r)}{R}$ (C) $\frac{v(R-r)}{R}$ (D) $\frac{vR}{R+r}$

29. The angular speed of bar as a function of θ is :

- (A) $\left(\frac{2v}{R+r}\right) \frac{\sin^2\left(\frac{\theta}{2}\right)}{\cos\left(\frac{\theta}{2}\right)}$ (B) $\left(\frac{v}{R+r}\right) \frac{1}{\sin^2\theta}$ (C) $\frac{2v}{R+r} \sin^2\left(\frac{\theta}{2}\right)$ (D) $\frac{v(R+r)}{R^2} \sin\theta$

PARAGRAPH FOR QUESTIONS 30 - 32

A disc of mass ' m ' and radius ' R ' is free to rotate in horizontal plane about a vertical smooth fixed axis passing through its centre. There is a smooth groove along the diameter of the disc and two small balls of mass $\frac{m}{2}$ each are placed in it on either side of the centre of the disc as shown in figure. The disc is given an initial angular velocity ω_0 and released.

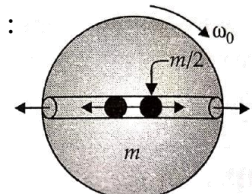


30. The angular speed the disc when the balls reach the end of the disc is :

- (A) $\frac{\omega_0}{2}$ (B) $\frac{\omega_0}{3}$ (C) $\frac{2\omega_0}{3}$ (D) $\frac{\omega_0}{4}$

31. The speed of each ball relative to the ground just after they leave the disc is :

- (A) $\frac{R\omega_0}{\sqrt{3}}$ (B) $\frac{R\omega_0}{\sqrt{2}}$
(C) $\frac{2R\omega_0}{3}$ (D) None of these

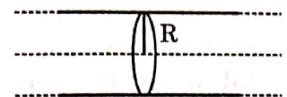


32. The net work done by forces exerted by the disc on each ball for the duration ball remains on the disc is :

- (A) $\frac{2mR^2\omega_0^2}{9}$ (B) $\frac{mR^2\omega_0^2}{18}$ (C) $\frac{mR^2\omega_0^2}{6}$ (D) $\frac{mR^2\omega_0^2}{9}$

PARAGRAPH FOR QUESTIONS 33 - 35

Consider a hypothetical planet which is very long and cylindrical. The density of the planet is ρ and its radius is R .



33. What is the possible orbital speed of the satellite in moving around the planet in circular orbit in a plane which is perpendicular to the axis of planet?

- (A) $R\sqrt{\pi G\rho}$ (B) $2R\sqrt{\pi G\rho}$ (C) $R\sqrt{2\pi G\rho}$ (D) $D\sqrt{\frac{G\rho}{2\pi}}$

34. If an object is projected radially outwards from the surface such that it reaches upto a maximum distance of $3R$ from the axis then what should be the speed of projection ?

- (A) $R\sqrt{\frac{2}{3}\pi G\rho}$ (B) $2R\sqrt{\pi G\rho \ln 3}$ (C) $R\sqrt{\frac{4}{3}\pi G\rho}$ (D) $R\sqrt{\frac{2}{3}\pi G \ln 3}$

35. Assume that the planet is rotating about its axis with time period T . How far from the axis of the planet do the geo synchronous tele-communications satellites orbit ?

- (A) $RT\sqrt{\pi G\rho}$ (B) $2RT\sqrt{\pi G\rho}$ (C) $RT\sqrt{2\pi G\rho}$ (D) $RT\sqrt{\frac{G\rho}{2\pi}}$

SECTION 3

MULTIPLE CORRECT ANSWER TYPE

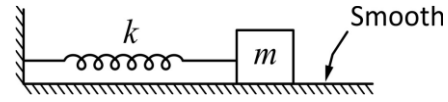
- *36. Figure shows an ideal spring block system, force constant of spring is k which has been compressed by an amount x_0 . If x is instantaneous deflection of spring from its natural length, mark the correct options.

(A) Instantaneous power developed by spring is $P = kx\sqrt{\frac{k}{2m}(x_0^2 - x^2)}$

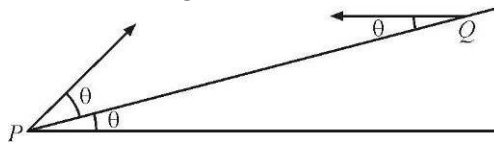
(B) Maximum power of spring is $\frac{k}{2}\sqrt{\frac{k}{m}}x_0^2$

(C) Maximum power of spring is $x = \frac{x_0}{\sqrt{2}}$

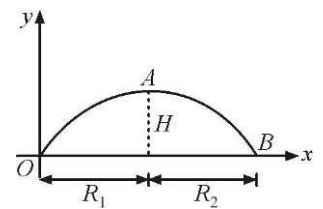
(D) Maximum power occurs at $x = \frac{x_0}{2}$



- *37. From an inclined plane two particles P , Q are projected with same speed at same angle θ , one up and other down the plane as shown in figure. Which of the following statement(s) is/are correct?

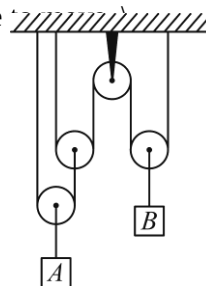


- (A) The particles will collide the plane with same speed
 (B) The times of flight of each particle are same
 (C) Both particles strike the plane perpendicularly
 (D) The particles may collide in mid-air if projected simultaneously and time of flight of each particle is less than the time of collision
- *38. In a projectile motion let $t_{OA} = t_1$ and $t_{AB} = t_2$. The horizontal displacement from O to A is R_1 and from A to B is R_2 . Maximum height is H and time of flight is T . If air drag is to be considered, then choose the correct alternative(s).
- (A) t_1 will decrease while t_2 will increase
 (B) H will increase
 (C) R_1 will decrease while R_2 will increase
 (D) T may increase or decrease



- *39. In figure shown if $m_A = 20$ kg and $m_B = 80$ kg. If system is set free

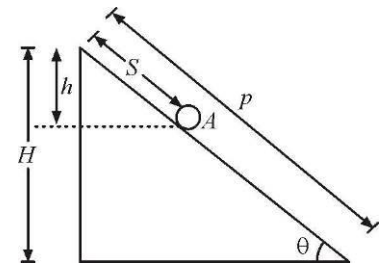
- (A) The acceleration of block A is $\frac{7g}{17}$
 (B) The acceleration of block B is $\frac{7g}{17}$
 (C) The acceleration of block A is $\frac{7g}{14}$
 (D) The acceleration of block B is $\frac{7g}{14}$



- *40. The potential energy of a particle moving along x -axis is given by $U = 20 + 5\sin(4\pi x)$, where U is in J and x is in metre under the action of conservative force:
- (A) If total mechanical energy is $20 J$, then at $x = 7/8 m$, particle is at equilibrium
 (B) If total mechanical energy is $20 J$, then at $x = 7/8 m$, particle is not at equilibrium
 (C) If total mechanical energy is $20 J$, then at $x = 3/8 m$, particle is at equilibrium
 (D) If total mechanical energy is $20 J$, then at $x = 3/8 m$, particle is not at equilibrium

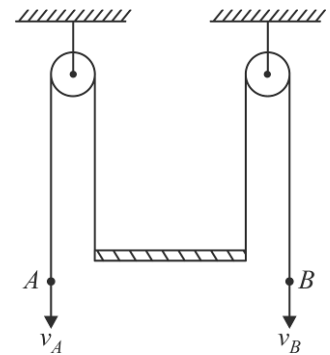
- *41. A particle slides down from rest on an inclined plane of angle θ with horizontal. The distances are as shown. The particle slides down to the position A, where its velocity is v .

- (A) $(v^2 - 2gh)$ will remain zero
 (B) $(v^2 - 2gs \sin \theta)$ will remain zero
 (C) $\left[v^2 - \frac{2gs(H-h)}{(p-s)} \right]$ will remain zero
 (D) $\left[v^2 - \frac{2gsH}{P} \right]$ will remain zero

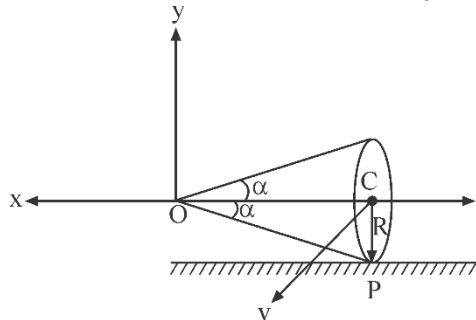


- *42. A thin uniform rod of mass 5 kg and length 1 m is held in horizontal position with the help of strings attached to ends of rod. Other ends of strings are held by some external agent. Now end A is pulled down with speed $v_A = 3t$ and B is pulled down with speed $v_B = t$ where t is time in sec. Choose the correct choices at time $t = 0$.

- (A) Angular acceleration of rod is 2 rad/s^2
 (B) Tension in the left string is $185/6 \text{ N}$
 (C) Acceleration of rod is 1 m/s^2
 (D) Tension in the right string is $170/3 \text{ N}$



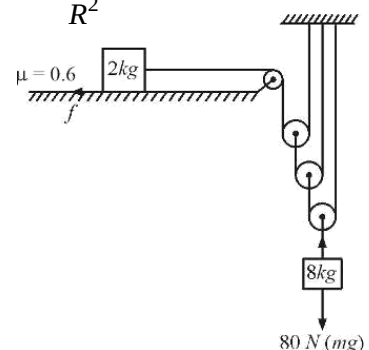
- *43. A round cone with half-angle α and the radius of the base R rolls uniformly and without slipping over a horizontal plane as shown in Figure. The cone apex is hinged at the point O which is on the same level with the point C, the cone base centre. The velocity of point C is V .



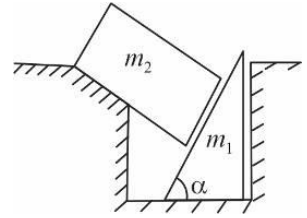
- (A) The magnitude of vector of the angular velocity of the cone is $\frac{V}{R \cos \alpha}$
 (B) The magnitude of vector of the angular acceleration of the cone is $\frac{V^2}{R^2} \tan \alpha$
 (C) The magnitude of vector of the angular velocity of the cone is $\frac{V}{R \sin \alpha}$
 (D) The magnitude of vector of the angular acceleration of the cone is $\frac{V^2}{R^2} \cot \alpha$

- *44. Two blocks of masses 2 kg and 8 kg are connected through pulleys and strings as shown in the figure. If all pulleys and strings are ideal, then: ($\mu = 0.6$, Take $g = 10 \text{ m/sec}^2$)

- (A) Friction force between the 2 kg block and the surface is 12 N
 (B) Friction force between the 2 kg block and the surface is 10 N
 (C) Acceleration of the 2 kg block is equal to zero
 (D) Tension in the string connected to the 2 kg block is 10 N

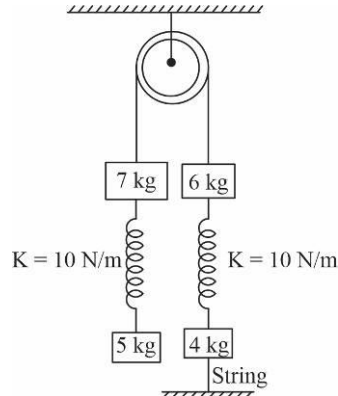


- *45. A wedge of mass m_1 and a block of mass m_2 are in equilibrium as shown. Inclined surface of the wedge has an inclination α with the horizontal. Each surface is frictionless. The normal reaction on the wedge may be:



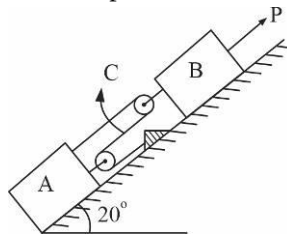
- (A) $m_2 g \cos \alpha$ (B) $m_2 g \sin \alpha \cos \alpha$
(C) $m_1 g + m_2 g \cos^2 \alpha$ (D) $m_1 g + m_2 g \sin \alpha \cos \alpha$

- *46. In the given diagram, pulley is frictionless and massless. Both the springs are having same force constant 10 N/m . Initially with the string attached to the grounds, the total system is at rest. Now if the string is cut, then immediately after cutting the string,



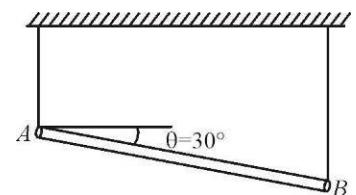
- (A) Acceleration of 7kg block is zero (B) Acceleration of 5kg block is zero
(C) Acceleration of 6kg block is $100/6 \text{ m/s}^2$ (D) Acceleration of 4kg block is 5 m/s^2

- *47. Under the action of force P , the constant acceleration of block B is 6 m/sec^2 up the incline. For the instant when the velocity of B is 3 m/sec up the incline. Choose the correct option(s)



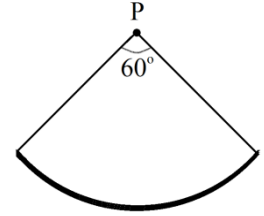
- (A) Velocity of B relative to A is 1 m/s
(B) Acceleration of B relative to A is 2 m/s^2
(C) The velocity of point C of the cable (in ground frame) is 4 m/s
(D) Velocity of B relative to A is 2 m/s

- *48. Figure shows a thin rod (mass M , length L) suspended in a vertical plane by tying ideal strings at ends A and B. Initially, rod makes an angle 30° with horizontal and is at rest. Now: the string connected to the end B is cut. Let T be the tension in the string connected to A be T and a_A be the magnitude of acceleration of point A just after the "cutting" event. Pick correct option (s):



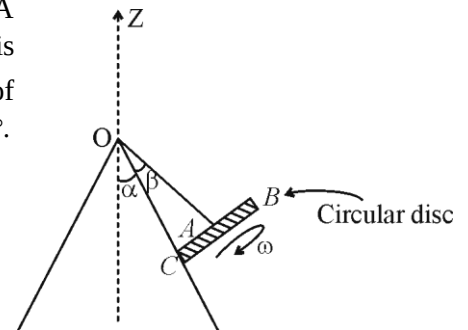
- (A) $T = \frac{3mg}{13}$ (B) $a_A = \frac{3g}{13}$ (C) $T = \frac{4mg}{13}$ (D) $a_A = \frac{3\sqrt{3}g}{13}$

- *49. A thin, uniform ring of radius $\frac{\pi}{4}$ m and mass $8\sqrt{3}$ kg is fixed with two rods of length $\frac{\pi}{4}$ m each. The system is pivoted about the centre of the ring P and made to rotate in the vertical plane. When the ring is in its lowest position (i.e. the position shown in the figure), the angular velocity of the system is 4 rad/s.
[Assume that the force on the ring by a rod is always purely along the length of the rod]
[$g = 10 \text{ m/s}^2$]



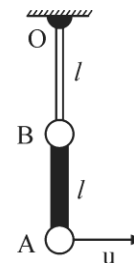
- (A) At this instant, the force exerted by one of the rods on the ring is 144 N.
(B) The centre of mass of any circular arc of uniform mass per unit length is at a distance $\frac{3R}{\pi}$ from the centre of the ring
(C) At this instant, the force exerted by one of the rods on the ring is 172 N.
(D) The centre of mass of any circular arc of uniform mass per unit length is at a distance $\frac{4R}{\pi}$ from the centre of the ring

- *50. A thin circular disc of mass m is rigidly fixed from its center A to point O via a massless rigid rod. Length $OC = l$. The disc is rolling over the conical surface, such that angular velocity of disc about its C.O.M. is given as ω (as shown). $\alpha = 30^\circ$, $\beta = 45^\circ$. Which of the following statements is / are true?



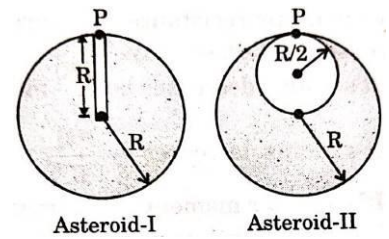
- (A) The center of mass of disc rotates about z-axis with angular speed of 2ω .
(B) Angular momentum of disc about its center of mass is $\frac{ml^2\omega}{4}$
(C) Torque due to external force about point O is $\frac{ml^2\omega^2}{8}(\sqrt{3}-1)$.
(D) Z-component of angular momentum about C.O.M. is $\frac{(\sqrt{3}-1)}{4\sqrt{2}}ml^2\omega$

- *51. A dumbbell of two masses A and B each of mass m joined by a light rod of length l with B attached to a suspension point O by a light rod of length l as shown in figure.



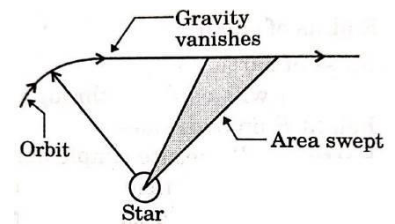
- (A) The minimum required speed with which mass A to be projected horizontally so that the system completes the vertical circular motion is $4\sqrt{\frac{3gl}{5}}$.
(B) The minimum required speed with which mass A to be projected horizontally so that the system completes the vertical circular motion is $2\sqrt{\frac{3gl}{5}}$.
(C) Velocity of the two masses will be Zero at the top position
(D) Tangential Velocity of B is half then A

- *52. Consider two solid spherical asteroid of uniform density of mass M and radius R . In one asteroid a tunnel of very small size of depth R is bored to the centre and in other asteroid a spherical cavity of radius $\frac{R}{2}$ is made as shown in the figure. Now, identical particles of mass m dropped into the cavities of both asteroids from the top most point P .

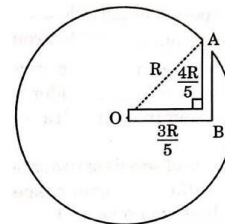


If force experienced by particle is F_I and F_{II} respectively in cavities of asteroids I and II, when they are x distance away from the centre of asteroids. If the time taken by particles to reach the centre of asteroids is T_I and T_{II} respectively then :

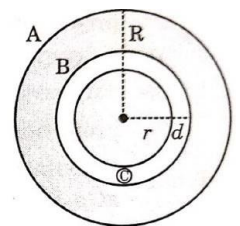
- (A) the ratio of $\frac{F_I}{F_{II}}$ is equal to $\frac{2x}{R}$ (B) the ratio of $\frac{F_I}{F_{II}}$ is equal to $\frac{x}{R}$
 (C) the ratio of $\frac{T_I}{T_{II}}$ is equal to $\frac{\pi}{4}$ (D) the ratio of $\frac{T_I}{T_{II}}$ is equal to $\frac{\pi}{2}$
- *53. A planet is orbiting a star when for no apparent reason the star's gravity suddenly vanishes. After which planet moves in a straight line. Mark the correct statement(s) :
- (A) Newton's first law is obeyed on planet after gravity vanishes
 (B) Kepler's law of areas is obeyed only till be planet is in gravity of star
 (C) Kepler's law of areas is obeyed even after gravity vanishes
 (D) Angular momentum of planet about centre of star is conserved through out its motion



- *54. A narrow smooth tunnel of L-shapes is made into earth up to centre as shown in the figure. A small ball which just fits in tunnel is released from 'A'. The collision of ball is perfectly inelastic. (g is acceleration due to gravity on surface of planet) :



- (A) time to fall from A to B is $\frac{\pi}{2}\sqrt{\frac{R}{g}}$ (B) time to fall from B to O is $\frac{\pi}{4}\sqrt{\frac{R}{g}}$
 (C) speed just before hitting B is $\frac{4R}{5}\sqrt{\frac{g}{R}}$ (D) speed just before hitting O is $\frac{R}{5}\sqrt{\frac{g}{R}}$
- *55. 'A' is a huge planet with uniform mass distribution, mass M , radius R . B is a circular tunnel made in it, concentric with planet. Radius of tunnel is r , $\left(r = \frac{R}{2}\right)$ cross-sectional diameter is d ($d \ll r$). C is a small ball of mass m ($m \ll M$) which moving freely inside the tunnel without friction in a uniform circular motion, take acceleration due to gravity on surface of A as g :



- (A) If contact force on C is double the gravitational force, its time period $= 2\pi\sqrt{\frac{R}{3g}}$.
 (B) If contact force on C is double the gravitational force, its time period $= 2\pi\sqrt{\frac{R}{g}}$.
 (C) If contact force is zero, time period T , radius of tunnel r , then $T \propto r$.
 (D) If contact force is zero, time period T , radius of tunnel r , then T would be independent of r .

SECTION 4

COLUMN MATCHING TYPE

Direction for Q. 56 -57

Two bodies (Body I and Body II) are released from rest at the top of an inclined plane in separate experiments. In both cases, the centre of the respective body is initially at a height h above the ground. Consider the following match list. In List-I, the symbols represent the following quantities for Body I and Body II:

I_1, I_2 Moment of inertia about the axis of the body

t_1, t_2 Time taken to reach the bottom of the inclined plane

v_1, v_2 Velocity of the CM upon reaching the bottom of the inclined plane

L_1, L_2 Angular momentum about the CM upon reaching the bottom of the inclined plane

Column-I		Column-II	
(P)	$\frac{I_1}{I_2}$	(i)	1
(Q)	$\frac{t_1}{t_2}$	(ii)	2
(R)	$\frac{v_1}{v_2}$	(iii)	$\sqrt{\frac{12}{13}}$
(S)	$\frac{L_1}{L_2}$	(iv)	$\sqrt{\frac{13}{12}}$
		(v)	$\frac{1}{2}$
		(vi)	$\sqrt{\frac{52}{75}}$
		(vii)	$\sqrt{\frac{75}{52}}$
		(viii)	$\frac{5}{4}$

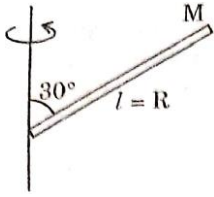
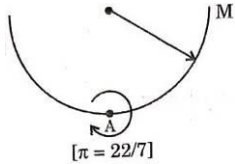
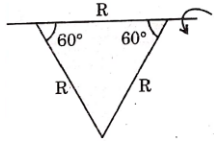
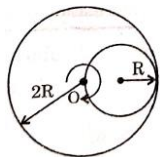
56. Body I is a uniform disc of mass m and radius R , and Body II is a uniform disc of mass $4m$ and radius $\frac{R}{2}$. The option that correctly matches List-I to List-II is:

(A) (p) – (i), (q) – (v), (r) – (i), (s) – (i) (B) (p) – (i), (q) – (i), (r) – (i), (s) – (ii)
 (C) (p) – (i), (q) – (i), (r) – (i), (s) – (v) (D) (p) – (i), (q) – (ii), (r) – (i), (s) – (i)

57. Body I is a uniform annular disc of mass m , inner radius $\frac{R}{2}$ and outer radius R , and Body II is a hollow cylinder of mass m and radius R . The option that correctly matches List-I to List-II is:

(A) (p) – (viii), (q) – (iv), (r) – (iii), (s) – (vii)
 (B) (p) – (ii), (q) – (vi), (r) – (vii), (s) – (iii)
 (C) (p) – (viii), (q) – (iii), (r) – (iv), (s) – (vii)
 (D) (p) – (ii), (q) – (vii), (r) – (vi), (s) – (iii)

58. Match the following columns :

Column-I		Column-II	
(A)	 Uniform Rod. M.I. about axis is	(P)	$\frac{8MR^2}{11}$
(B)	 Uniform Semiconductor Ring. Axis is perpendicular to plane of ring passing A.	(Q)	$\frac{MR^2}{12}$
(C)	 Uniform triangular plate of mass M.	(R)	$\frac{13MR^2}{8}$
(D)	 Uniform disk of initial mass M from which circular portion of radius R is then removed. M.I. of remaining mass about axis which is perpendicular to plane of plate.	(S)	$\frac{MR^2}{8}$

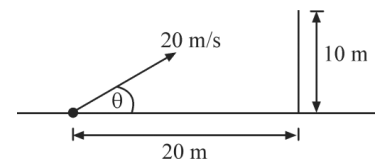
59. A particle is projected vertically up from the surface of the earth with an speed $\sqrt{gR}$.

Column-I		Column-II	
(A)	Maximum height attained by the particle is αR then α is	(P)	2
(B)	Time taken by the particle to reach the maximum height is $\beta \sqrt{\frac{R^3}{GM}}$ then β is	(Q)	$\frac{\pi}{4} + 1$
(C)	Value of acceleration due to gravity at maximum height is $\gamma \frac{GM}{4R^2}$ then γ is	(R)	$\frac{\pi}{2} + 1$
(D)	Orbital velocity of a satellite revolving at maximum height is $\delta \sqrt{\frac{GM}{2R}}$ then δ is	(S)	1
		(T)	None of these

60. A satellite is to be launched and established in its orbit around the Earth. For this purpose it is raised to a certain height (equal to its orbit distance) with a rocket and held stationary by some mechanism and an explosion is carried out instantly to provide the required velocity to the satellite. Let the original mass of satellite is M_s , mass of the earth M_e , radius of the required orbit R and universal gravitational constant G . As a result of explosion the satellite breaks into two parts of masses m_1 and m_2 and energy released is given by E . Column-I gives certain combination of m_1 , m_2 and E while column-II gives the nature of orbit followed by m_2 . Match them suitably (If $v > v_e$; orbit is hyperbolic, $v = v_e$; orbit is parabolic and $v < v_e$; orbit is elliptical where v_e is escape velocity and v is velocity of the part) :

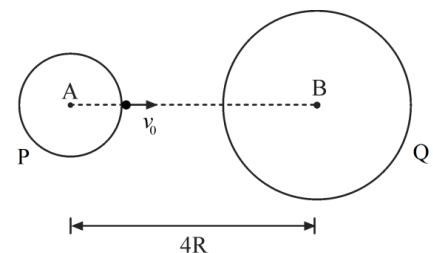
Column-I		Column-II	
(A)	$m_1 = m_2; E = \frac{GM_e M_s}{R}$	(P)	Parabolic orbit
(B)	$m_1 = 2m_2; E = \frac{GM_e M_s}{4R}$	(Q)	Hyperbolic orbit
(C)	$m_1 = 3.5m_2; E = \frac{GM_e M_s}{4R}$	(R)	Circular orbit
(D)	$m_1 = \frac{1}{3}m_2; E = \frac{2GM_e M_s}{R}$	(S)	Elliptical orbit

61. A ball is thrown from a point on the ground at an angle θ with the horizontal towards a fixed vertical wall of height 10 m as shown. The initial velocity of the ball is 20 m/s. Match the columns given below. [$\tan 15^\circ = 2 - \sqrt{3}$, $\tan 75^\circ = 2 + \sqrt{3}$, $g = 10 \text{ m/s}^2$]



Column-I		Column-II	
(A)	The ball falls between the point A and the wall if	(P)	$45^\circ < \theta < \tan^{-1}(3)$
(B)	The ball hits the wall if	(Q)	$15^\circ < \theta < 45^\circ$
(C)	The ball goes beyond the wall if	(R)	$\tan^{-1}(3) < \theta < 75^\circ$
		(S)	$\theta < 15^\circ$ OR $\theta > 75^\circ$
		(T)	$15^\circ < \theta < 45^\circ$ OR $\tan^{-1}(3) < \theta < 75^\circ$

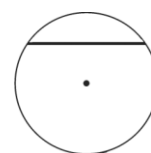
62. Two homogeneous spheres P and Q of radius R and $2R$ and mass M and $4M$ are fixed with their centres at the points A and B as shown. A small particle of mass m is launched directly towards B, from a point on the surface of P as shown with initial velocity v_0 directed along AB. Let the minimum value of v_0 such that the particle is able to reach the surface of Q be $v_{\min}$. Match the columns given below. The



quantities in Column II are in units of $\sqrt{\frac{GM}{R}}$.

Column-I		Column-II	
(A)	$v_{\min}$	(P)	$\frac{1}{\sqrt{3}}$
(B)	If $v_0 = v_{\min}$, the velocity with which the particle reaches the surface of Q	(Q)	$\frac{1}{\sqrt{6}}$
(C)	If $v_0 = v_{\min}$, the velocity of the particle when it reaches the mid-point of AB	(R)	$\frac{5}{\sqrt{6}}$
		(S)	$\frac{1}{\sqrt{2}}$
		(T)	$\sqrt{\frac{2}{3}}$

63. A rod of mass m and length $\sqrt{3}R$ is fixed rigidly along a chord of a massless disc of radius R . The disc is in equilibrium on a rough horizontal surface with the rod parallel to the surface as shown. Now, the disc is given a small push such that it starts rolling without slipping on the surface at $t = 0$. At $t = t_1$, the rod becomes vertical for the first time, and at $t = t_2$, the rod becomes horizontal again for the first time. Match the columns below. The quantities in column II are in units of $\sqrt{gR}$.



Column-I		Column-II	
(A)	At $t = t_1$, the velocity of the centre of the disc is	(P)	$\frac{1}{\sqrt{6}}$
(B)	At $t = t_1$, the velocity of the centre of the rod is	(Q)	1
(C)	At $t = t_2$, the velocity of the centre of the disc is	(R)	$\sqrt{\frac{2}{3}}$
(D)	At $t = t_2$, the velocity of the centre of the rod is	(S)	$\sqrt{\frac{5}{6}}$
		(T)	2
		(U)	$\frac{\sqrt{5}}{2}$

64. A uniform, thin rod AB of length L is moving in the X-Y plane such that at a particular instant, the velocity of its ends are $\vec{v}_A$ and $\vec{v}_B$. For the same instant, match the following columns.

Column-I		Column-II	
(A)	The velocity of the centre of the rod is	(P)	$\vec{v}_A - \vec{v}_B$
(B)	The angular velocity of the rod is	(Q)	$\sqrt{\frac{ \vec{v}_A ^2 + \vec{v}_B ^2}{2}}$
(C)	A vector perpendicular to the length of the rod is	(R)	$\frac{\vec{v}_A + \vec{v}_B}{2}$
		(S)	$\sqrt{ \vec{v}_A \vec{v}_B }$
		(T)	$\frac{ \vec{v}_A + \vec{v}_B }{2L}$
		(U)	$\frac{ \vec{v}_A - \vec{v}_B }{L}$

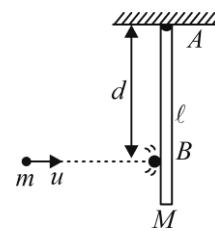
65. In gravity free space, a charged particle is moving in a helical path of constant pitch and constant time period T under the influence of a uniform magnetic field. The axis of the helix is a line in the X-Y plane making an acute angle θ with the positive Y-axis. At $t=0$, the particle is at the origin and its velocity is $\vec{v} = v_0 \hat{j}$. Match the following columns for subsequent times.

Column-I		Column-II	
(A)	The X-coordinate of the particle is given by	(P)	$v_0 \left((\cos^2 \theta) t + \left(\frac{T}{2\pi} \sin^2 \theta \right) \sin \left(\frac{2\pi}{T} t \right) \right)$
(B)	The Y-coordinate of the particle is given by	(Q)	$v_0 \sin \theta \cos \theta \left(t - \left(\frac{T}{2\pi} \right) \sin \left(\frac{2\pi}{T} t \right) \right)$
(C)	The Z-coordinate of the particle is given by	(R)	$v_0 \left(\cos^2 \theta + \left(\sin^2 \theta \right) \cos \left(\frac{2\pi}{T} t \right) \right)$
(D)	The X-component of the particle's velocity is given by	(S)	$v_0 \sin \theta \cos \theta \left(1 - \cos \left(\frac{2\pi}{T} t \right) \right)$
(E)	The Y-component of the particle's velocity is given by	(T)	$v_0 \sin \theta \sin \left(\frac{2\pi}{T} t \right)$
(F)	The Z-component of the particle's velocity is given by	(U)	$(v_0 \sin \theta) \left(\frac{T}{2\pi} \right) \left(1 - \cos \left(\frac{2\pi}{T} t \right) \right)$

SECTION 5

NUMERICAL TYPE

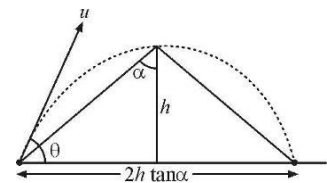
66. A uniform rod AB of mass M and length ℓ is hinged at point A and it is hanging freely to rotate in a vertical plane. A ball of mass m moving horizontally hits it at a distance d from hinge as shown. The d for which during collision hinge will not exert any impulse on rod is $n \times \ell$. Find n .



67. A sailor in a boat, which is going due east with a speed of 8 m/s , observes that a submarine is heading towards north at a speed of 12 m/s and sinking at a rate of 2 m/s . The commander of submarine observes a helicopter ascending at a rate of 5 m/s and heading towards west with 4 m/s . Find the actual speed of the helicopter.

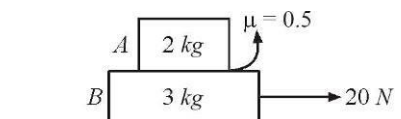
68. Three particles A , B and C situated at vertices of an equilateral triangle all moving with same constant speed such that A always move towards B and B always towards C and C always towards A . Initial separation between each of particle is a . O is the centroid of the triangle. Distance covered by particle A when it completes one revolution around O is $\frac{2a}{k}(1 - e^{-\sqrt{3}\pi})$ find value of k .

69. A projectile is fired from the base of cone-shaped hill. The projectile grazes the vertex and strikes the hill again at the base. If α be the half-angle of the cone, h its height, u the initial velocity of projection and θ angle of projection, then $\tan\theta \tan\alpha$ is.

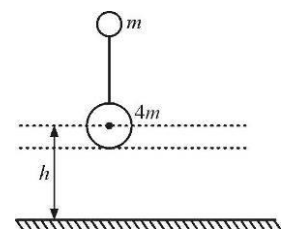


70. A heavy particle is projected from a point at the foot of a fixed plane, inclined at an angle 45° to the horizontal. If $\phi (> 45^\circ)$ is the inclination to the horizontal of the initial direction of projection, for what value of $\tan\phi$ will the particle strike the plane horizontally.

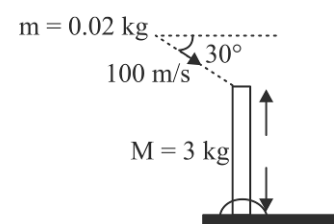
71. Two blocks A and B are placed one over other. Block B is acted upon by a force of 20 N which displaces it through 5 m . Find work done (in Joules) by frictional force on block A .



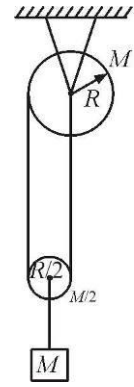
72. A small ball of mass ' m ' is connected by an inextensible massless string of length ' $l = 10 \text{ m}$ ' with an another ball of mass $M = 4 \text{ m}$. They are released with zero tension in the string from a height h ($h = 5 \text{ m}$) as shown. Find the time when the string becomes taut for the first time after the mass ' M ' collides with the ground is _____ S. (Take all collisions to be elastic and time during collision to be negligible) ($g = 10 \text{ m/s}^2$)



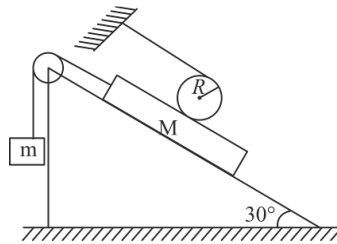
73. A bullet of mass 0.02 kg is travelling with speed 100 m/s at an angle of 30° with respect to the horizontal. The bullet collides with and sticks to the end of a vertical thin rod of length 2.0 m and mass 3 kg as shown in the figure. The rod then rotates about its end which is hinged to the floor. Find the angular velocity of the rod immediately after the collision.



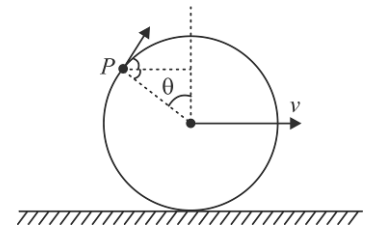
74. A cylindrical pulley of mass M and radius R is connected to another cylindrical pulley of mass $M/2$ and radius $R/2$ through a massless string as shown in figure. If sufficient string is wrapped over larger pulley, find the acceleration of mass M attached to the smaller pulley, when whole system is released from rest. (Assume there is no slipping in the pulley)



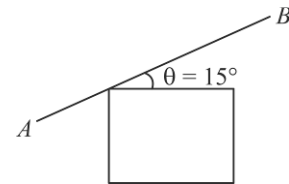
75. A plank of mass M is placed over smooth triangular wedge with angle of inclination 30° . Block of mass m is connected to the plank by massless string. A cylinder of mass M and radius R is now placed over plank, such that one end of the string, wrapped over cylinder is fixed and parallel to incline. Coefficient of friction between cylinder and plank is sufficient for pure rolling of cylinder. If system is released from rest with condition that $M = 4m$, the acceleration of plank is $\frac{kg}{13}$ find k ?



76. A wheel of radius $R = 2m$ performs pure rolling on a rough horizontal surface with speed $V = 10$ m/s. In the figure shown angle θ is angular position of point P on wheel from where dirt detaches from wheel and reaches the maximum height from ground. Find the value of $\sec\theta$. (Take $g = 10$ m/s²)

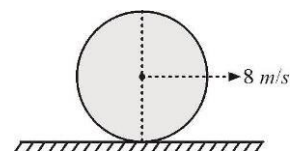


77. A square plate has uniform mass distribution. It has mass $M = 24/11$ kg and edge $L = 2m$ as shown in the figure. Calculate the moment of inertia of the plate about the axis AB (as shown in figure) in the plane of the square plate. (in SI unit)



78. A particle starts moving from rest such that it has uniform acceleration a_0 during the 1st second of motion, uniform acceleration $\frac{a_0}{2}$ during the 2nd second of motion, uniform acceleration $\frac{a_0}{4}$ during the 3rd second of motion, and so on, such that it has uniform acceleration $\frac{a_0}{2^{n-1}}$ during the n^{th} second of motion. If the displacement of the particle during the 5th second of motion is $D_5 = \frac{Ka_0}{32}$, the value of K is _____.

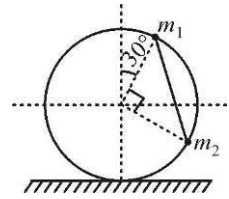
79. A 10 kg solid sphere of radius $r = 0.8$ m is rolling without slipping on a horizontal rough surface with 8 m/s. The force applied by the right half of the sphere on the left half is $30x$ Newton. Find the value of x .



80. A particle is tied to a massless string of length 20 cm and the other end of the string is fixed. Now the particle is made to move in a vertical circle such that the ratio of its maximum and minimum velocity

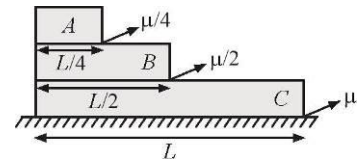
is $\frac{v_{\max}}{v_{\min}} = \sqrt{2}$. Then, $v_{\max} = \underline{\hspace{2cm}}$ m/s. $[g = 10 \text{ m/s}^2]$

81. Two beads of masses m_1 and m_2 are connected to each other by massless string. Both beads are free to move in fixed smooth circular loop in vertical plane. At any instant they are in the position shown and subtends angle 90° at centre. Find tension in string if string remains taut and if $T = \frac{(\sqrt{3}-1)m_1m_2g}{\sqrt{k(m_1+m_2)}}$. Find k ?

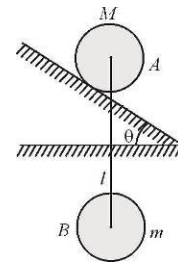


82. A long plank begins to move at $t = 0$ and accelerates along a straight track with a speed given by $v = 2t^2$ for $0 \leq t \leq 2$ (where v is in m/s and t is in second). After 2 sec the plank continues to move at the constant speed acquired. A small block initially at rest on the plank begins to slip at $t = 1$ sec and stops sliding at $t = 3$ sec. If the coefficient of static friction and kinetic friction between the plank and the block is 0.s and 0.k (where s and k are digits) respectively, find $s + k$. (Take $g = 10 \text{ m/s}^2$)

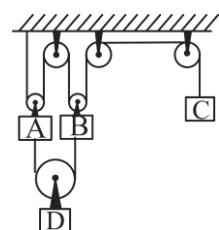
83. Three blocks A, B and C of mass m , $\frac{m}{2}$ and m of different densities and dimensions are placed over each other as shown in the figure. The coefficients of friction are shown. Blocks placed in a vertical line are made to move towards right with same velocity at the same instant. Find the time (in sec) taken by the upper block A to topple from the middle block B. Assume that blocks B and C don't stop sliding before A topples from B. (Given $L = 36 \text{ m}$, $\mu = 0.4$ and $g = 10 \text{ m/s}^2$)



84. A disc A of mass M is placed at rest on the smooth incline surface of inclination θ . A ball B of mass m is suspended vertically from the centre of the disc A by a light inextensible string of length l as shown in the figure. If the acceleration of the disc B immediately after the system is released from rest is $\frac{(M + km)g \sin \theta}{M + m \sin^2 \theta}$. Find k .



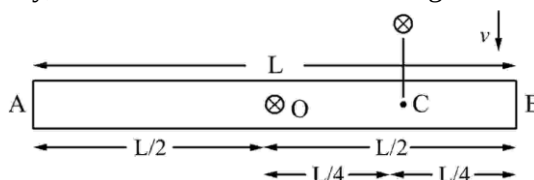
85. The system shown in figure starts from rest, and each block moves with a constant acceleration. If the relative acceleration of block C with respect to block B is 6 m/s^2 upwards and the relative acceleration of block D with respect to block A is 11 m/s^2 downwards, determine the velocity of block C after 3 s from start.



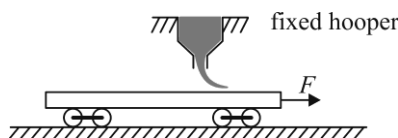
SECTION 6

SUBJECTIVE TYPE

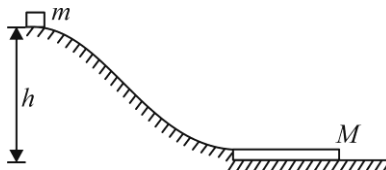
86. A homogeneous rod of length, $L = 1.8\text{m}$ and mass $M\text{ kg}$, is pivoted at the centre O , in such a way, that it can rotate freely in the vertical plane (see figure). The rod is initially in the horizontal position. A insect of mass M (same as the rod) falls vertically, with speed v , on the point C , at a distance $\frac{L}{4}$ from one end of the rod. The insect moves towards the closer end, immediately after falling on C . Consequently, the rod rotates with constant angular velocity ω .



- Determine the constant angular velocity (ω) in terms of v and L .
 - if the body reaches the end (B), when the body has turned through an angle of 90° , find v (m/s).
87. Figure shows a fixed hooper through which sand spills on a flat rail road car at a constant rate of $r\text{ kg/sec}$. Car is pulled on smooth surface by a constant force F . If mass of car is m_0 , find :

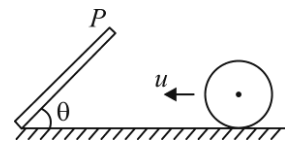


- The time function of velocity of car.
 - The time function of acceleration of car.
88. A spaceship of mass m_0 moves in the absence of external forces with a constant speed v_0 . To change the direction of motion, gases are ejected from rocket in a direction normal to its motion at velocity u relative to spaceship such that it moves in a circle of constant radius. Find the angle by which the direction of motion of spaceship changes when its mass decreases to m .
89. A particle of mass m_1 collides elastically with a stationary particle of mass m_2 ($m_1 > m_2$). Find the maximum angle through which the striking particle may deviate as a result of collision.
90. A particle of mass m_1 experienced a perfectly elastic collision with a stationary particle of mass m_2 . Find the fraction of kinetic energy lost by striking particle if it recoils at right angle to original direction of motion.
91. Figure shows a small body of mass m released from a height h on a smooth hill and at bottom of hill the body reaches on a plank of mass M placed on smooth ground. Due to friction between body and plank body slows down and after some time moves as one piece with plank. Find the work done by friction forces in this process of motion.

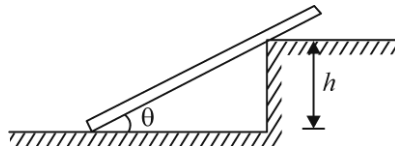


92. A sphere of mass m slides with velocity u on a smooth horizontal surface on which a rigid plank P is fixed at an angle θ as shown. If collision of sphere with plank is elastic find:

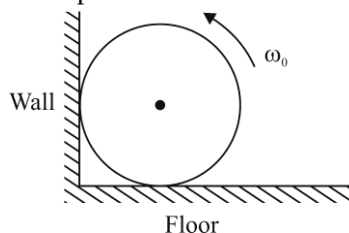
- (i) The impulse by plank on sphere during impact.
(ii) The impulse by floor on sphere during impact.



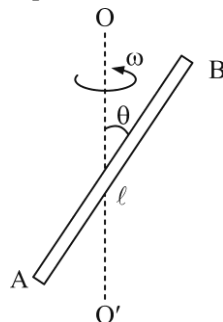
93. A uniform rod of length L rests against a step corner of height h as shown. Find the friction coefficient between ground and lower end of rod if the minimum angle the rod can rest on ground is θ . There is no friction between rod and corner of step.



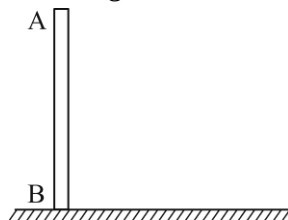
94. A uniform cylinder of radius R is spun about its axis at angular speed ω_0 and then placed into a corner. The coefficient of friction between cylinder, wall and floor is μ . Find the total number of turns cylinder will make before it stops.



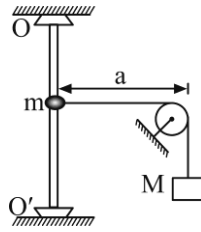
95. Figure shows a thin uniform rod AB of mass m and length ℓ rotating about the vertical axis OO' at angular velocity ω . Find the torque of centrifugal force acting on rod about its mid point C in the frame of reference rotating at angular speed ω attached to axis OO' .



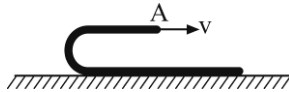
96. A thin uniform rod AB is vertically placed on a rough ground as shown. If its upper end is slightly tapped so that it starts falling and due to friction its lower end does not slip. Find the friction at B when rod is at an angle θ to vertical during fall.



97. A ring of mass m which can slide on a smooth vertical rod OO' is attached to a block of mass M ($M > m$) as shown in figure. If the system is released from rest from the state shown, find the distance fallen by ring before it comes to rest for the first time.



98. Figure shows a rope resting on rough ground. One end A of rope is pulled by an external force such that it moves at a uniform speed v . If mass per unit length of rope is $\lambda \text{ kg/m}$, find the power associated with the external force pulling the rope. Assume there is no slipping of rope on ground.



99. A body of mass m is projected on a rough horizontal surface with velocity v at $x = O$. The friction coefficient on surface is varying with distance at $\mu = bx$. Find the maximum instantaneous power of friction force during motion of body.

100. Figure shows a system of two bodies A and B of mass m each. Ground is smooth whereas the friction coefficient between A and B is μ . Initially spring is in its natural length l_0 . A force F slowly drags the bodies to a position where B starts sliding over A and it happens when spring makes an angle θ with its initial position with the vertical. Find work done by external force from start to this position when B starts sliding.

